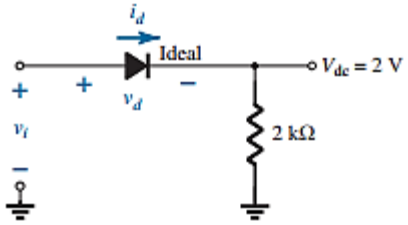
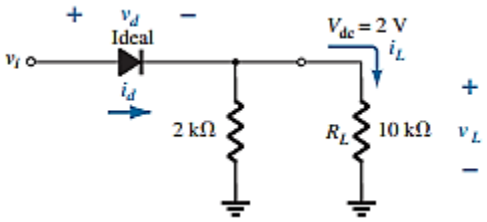
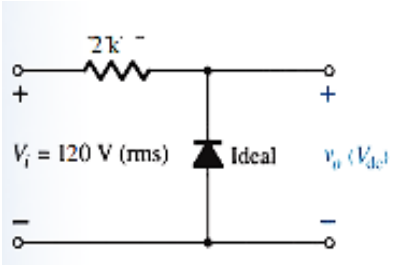
1 (a)Assuming an ideal diode, sketch v_i, v_d, and i_d for the half-wave rectifier below. The input is a sinusoidal waveform with a frequency of 60 Hz. Determine the profit value of v_i from the given dc level. (b)Repeat part (a) using silicon diode (V_K=0.7V).



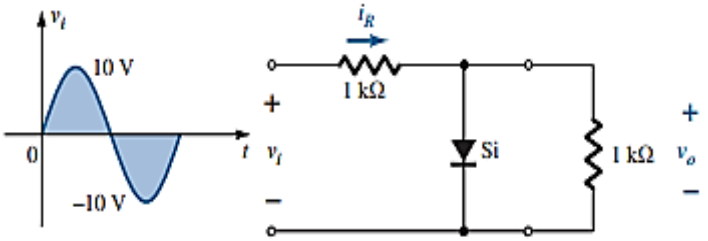
2 Repeat Problem (1) with a 10 kΩ load applied as shown in figure below. Sketch V_L and i_L.



3 For the network below, sketch v_o and determine V_{dc}.

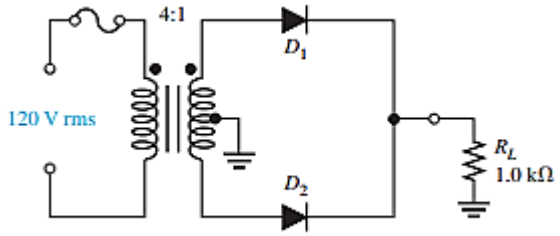


4 For the network of below, sketch V_o and i_R

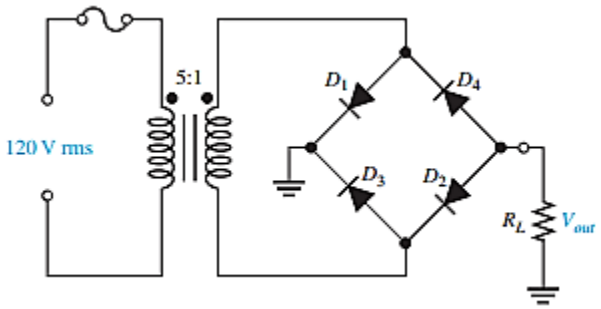


5 A full-wave bridge rectifier with a 120-V rms sinusoidal input has a load resistor of 1 kΩ.
a. If silicon diodes are employed, what is the dc voltage available at the load?
b. Determine the required PIV rating of each diode.
c. Find the maximum current through each diode during conduction.
d. What is the required power rating of each diode?

6 Consider the circuit below.
(a) What type of circuit is this?
(b) What is the total peak secondary voltage?
(c) Find the peak voltage across each half of the secondary.
(d) Sketch the voltage waveform across R_L.
(e) What is the peak current through each diode?
(f) What is the PIV for each diode?



7 Draw the output voltage waveform for the bridge rectifier shown below



8 Determine vo and the required PIV rating of each diode for the configuration below. In addition, determine the maximum current through each diode.

